Econ 101A
Section 12: Solutions

Clotaire Boyer*

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Until now we had one consumer and one firm. Today we build a market. There are three steps. First we add up individual supply and demand curves. Then we find the price that clears the market. Then we ask how that price moves when something changes. Elasticities let us answer the last question without worrying about units, and they also tell us who really pays a tax. Everything today is Lecture 9, first half.

1 Aggregation

Industry supply. With J firms, each with supply $y^j(p, w, r)$, industry supply is

$$Y^S(p, w, r) = \sum_{j=1}^{J} y^j(p, w, r).$$

Market demand. With J consumers, each with income M^j,

$$X^D(p_1, \ldots, p_n, M^1, \ldots, M^J) = \sum_{j=1}^{J} x^j(p_1, \ldots, p_n, M^j).$$

Add quantities, not prices. Fix a price. Ask each agent how much they want at that price. Add the answers. Graphically this is a *horizontal* sum of the curves. Two warnings. Each individual curve can have a kink where the agent starts or stops trading, so the sum has kinks too. And market demand depends on how income is split across people, not only on total income.

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2 Equilibrium in the Short Run

In the short run the number of firms J is fixed. The equilibrium price p^* sets excess demand to zero:

$$Y^S(p^*, w, r) = X^D(p^*, M^1, \ldots, M^J).$$

Nothing stops firms from making positive profits in the short run. That changes in the long run (Section 13).

3 Comparative Statics of Equilibrium

Let a parameter α shift supply, demand, or both. Write equilibrium as an implicit function

$$F(p, \alpha) \equiv Y^S(p, \alpha) - X^D(p, \alpha) = 0.$$

The implicit function theorem gives

$$\frac{dp^*}{d\alpha} = -\frac{F_\alpha}{F_p} = -\frac{\partial Y^S/\partial\alpha - \partial X^D/\partial\alpha}{\partial Y^S/\partial p - \partial X^D/\partial p}.$$

Supply slopes up and demand slopes down, so the denominator is positive. The sign of $dp^*/d\alpha$ is minus the sign of the numerator: anything that raises demand or cuts supply at a given price raises p^*.

4 Elasticities

The size of $dp^*/d\alpha$ depends on units (dollars or cents, tons or pounds). Elasticities remove the units:

$$\varepsilon_{x,p} = \frac{\partial x}{\partial p}\,\frac{p}{x} = \frac{\partial \ln x}{\partial \ln p}.$$

It is the percent change in x for a one percent change in p. Multiplying the comparative statics by α/p and using $X^D = Y^S$ at the equilibrium,

$$\varepsilon_{p,\alpha} = -\frac{\varepsilon_{S,\alpha} - \varepsilon_{D,\alpha}}{\varepsilon_{S,p} - \varepsilon_{D,p}}.$$

This is useful because empirical work gives us elasticities, not slopes.

5 Who Pays a Tax?

A per-unit tax t drives a wedge between the price buyers pay, p, and the price sellers keep, $p - t$. Equilibrium solves $Y^S(p - t) - X^D(p) = 0$. The implicit function theorem gives

$$\frac{\partial p^*}{\partial t} = \frac{\varepsilon_{S,p}}{\varepsilon_{S,p} - \varepsilon_{D,p}}, \qquad \frac{\partial (p^* - t)}{\partial t} = \frac{\varepsilon_{D,p}}{\varepsilon_{S,p} - \varepsilon_{D,p}}.$$

The two numbers are between 0 and 1 in absolute value and add up to one in absolute value. The side of the market that is *less* elastic pays more of the tax. Who hands the money to the government does not matter.

6 Exercises

Exercise 1 (building a market) Each of $J = 10$ identical firms has cost $c(y) = 4 + y^2$ if $y > 0$ and $c(0) = 0$. The 4 is a setup cost the firm avoids by shutting down. There are 5 identical consumers, each with demand $x(p) = \max\{20 - p,\, 0\}$.

(a) Find MC, AC, the minimum of AC, and the supply function of one firm.

(b) Find industry supply $Y^S(p)$. Sketch it. What happens at $p = 4$?

(c) Find market demand $X^D(p)$. A classmate instead adds up the inverse demands $p = 20 - x$ and writes $p = 100 - 5X$. What went wrong?

(d) Find the short-run equilibrium price, total quantity, output per firm, and profit per firm.

Solution.

(a) Marginal and average cost for $y > 0$:

$$\mathrm{MC} = c'(y) = 2y, \qquad \mathrm{AC} = \frac{4 + y^2}{y} = \frac{4}{y} + y.$$

Minimize AC:

$$\frac{d\,\mathrm{AC}}{dy} = -\frac{4}{y^2} + 1 = 0 \implies y^2 = 4 \implies y = 2, \qquad \mathrm{AC}(2) = \frac{4}{2} + 2 = 4.$$

Check: $\mathrm{MC}(2) = 2 \cdot 2 = 4 = \mathrm{AC}(2)$. MC cuts AC at its minimum, as in Section 11.

If the firm produces, it sets $p = \mathrm{MC} = 2y$, so $y = p/2$. It produces only if $p \geq \min \mathrm{AC} = 4$. Profit at $p = 4$ is $4 \cdot 2 - (4 + 4) = 0$, so the firm is indifferent there. Supply of one firm:

$$y(p) = \begin{cases} 0 & p < 4, \\ \dfrac{p}{2} & p \geq 4. \end{cases}$$

(b) Add the 10 firms at each price:

$$Y^S(p) = 10 \cdot y(p) = \begin{cases} 0 & p < 4, \\ 10 \cdot \dfrac{p}{2} = 5p & p \geq 4. \end{cases}$$

At $p = 4$ supply jumps from 0 to $5 \cdot 4 = 20$. Each firm jumps from 0 to 2 units, because it never wants to produce less than the output at minimum AC.

(c) At any $p \le 20$, each consumer buys $20 - p$. Five of them buy

$$X^D(p) = 5\,(20 - p) = 100 - 5p \quad (p \le 20), \qquad X^D(p) = 0 \quad (p > 20).$$

The inverse is $p = 20 - X/5$. The classmate summed *prices*, not quantities. That adds up the willingness to pay of five people for the *same* unit, which makes sense only for a public good. Quick check: at $p = 10$ each consumer buys 10, so the market buys 50. The classmate's curve at $X = 50$ gives $p = 100 - 250 = -150$. Clearly wrong.

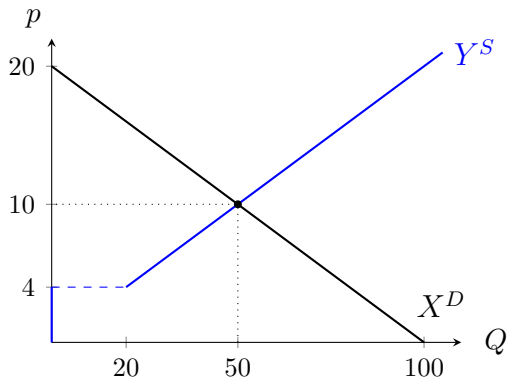
(d) Guess that $p^* \ge 4$, so we use the upward branch of supply:

$$5p = 100 - 5p \implies 10p = 100 \implies p^* = 10.$$

The guess holds since $10 \ge 4$. Quantity: $Y^* = 5 \cdot 10 = 50$, and check $X^D(10) = 100 - 50 = 50$. Each firm makes $y = 10/2 = 5$. Profit per firm:

$$\pi = p\,y - c(y) = 10 \cdot 5 - (4 + 5^2) = 50 - 29 = 21 > 0.$$

Firms make money in the short run. Keep this number: in Section 13 it will attract entry.



The dashed segment marks the jump at $p = 4$. No firm supplies a quantity between 0 and 2 units.

Exercise 2 (income distribution matters) Two consumers have Cobb-Douglas utility over good x and a composite good. Consumer j spends a share α_j of income M^j on x, so $x^j = \alpha_j M^j / p$.

(a) Find market demand for x.

(b) Let $\alpha_1 = 0.2$, $\alpha_2 = 0.6$ and total income $M^1 + M^2 = 1000$. Compare market demand when $(M^1, M^2) = (500, 500)$ and when $(M^1, M^2) = (800, 200)$.

(c) Under what condition does market demand depend only on total income?

(d) Supply of x slopes up. Income moves from $(500, 500)$ to $(800, 200)$. What happens to p^*? Use the formula for $dp^*/d\alpha$.

Solution.

(a) Add the two demands at a given price:

$$X^D(p) = \frac{\alpha_1 M^1}{p} + \frac{\alpha_2 M^2}{p} = \frac{\alpha_1 M^1 + \alpha_2 M^2}{p}.$$

(b) Equal split: $\alpha_1 M^1 + \alpha_2 M^2 = 0.2 \cdot 500 + 0.6 \cdot 500 = 100 + 300 = 400$, so $X^D = 400/p$.

Unequal split: $0.2 \cdot 800 + 0.6 \cdot 200 = 160 + 120 = 280$, so $X^D = 280/p$.

Same total income, but demand is 30% lower at every price. Money moved toward the person who spends less of it on x.

(c) Write $M^2 = M - M^1$ where M is total income. Then

$$X^D = \frac{\alpha_1 M^1 + \alpha_2 (M - M^1)}{p} = \frac{\alpha_2 M + (\alpha_1 - \alpha_2) M^1}{p}.$$

This does not depend on M^1 if and only if $\alpha_1 = \alpha_2$. Then $X^D = \alpha M / p$. The general idea: only total income matters when every consumer spends an extra dollar on x in the same way.

(d) Let the parameter be M^1 (with $M^2 = 1000 - M^1$). Supply does not depend on it, so $\partial Y^S / \partial M^1 = 0$. Demand does:

$$\frac{\partial X^D}{\partial M^1} = \frac{\alpha_1 - \alpha_2}{p} = \frac{0.2 - 0.6}{p} < 0.$$

Then

$$\frac{dp^*}{dM^1} = -\frac{0 - \partial X^D / \partial M^1}{\partial Y^S / \partial p - \partial X^D / \partial p} = \frac{\partial X^D / \partial M^1}{(+)} < 0.$$

The price of x falls. Redistribution is a demand shock for some goods and not for others.

Exercise 3 (comparative statics with the implicit function theorem) Go back to Exercise 1, but let the firm's cost be $c(y) = 4 + w\,y^2$ (so $w = 1$ before) and let demand be $X^D = 100 + \alpha - 5p$ (so $\alpha = 0$ before). Assume the price stays on the upward part of supply.

(a) Find the supply of one firm above its shutdown price. Write the equilibrium condition as $F(p, w, \alpha) = 0$.

(b) Find $dp^*/d\alpha$ with the implicit function theorem. Check it by solving for p^* directly.

(c) Find dp^*/dw and evaluate it at $w = 1$, $\alpha = 0$. Interpret the sign.

(d) At $w = 1$, $\alpha = 0$, compute $\varepsilon_{S,w}$, $\varepsilon_{D,w}$, $\varepsilon_{S,p}$, $\varepsilon_{D,p}$, and use the elasticity formula to get $\varepsilon_{p,w}$. Check with (c).

Solution.

(a) MC $= 2wy$. Setting $p = 2wy$ gives $y = p/(2w)$. Ten firms supply $Y^S = 10 \cdot p/(2w) = 5p/w$. Equilibrium:

$$F(p, w, \alpha) \equiv \frac{5p}{w} - (100 + \alpha - 5p) = \frac{5p}{w} + 5p - 100 - \alpha = 0.$$

(b) Partial derivatives:

$$F_p = \frac{5}{w} + 5, \qquad F_\alpha = -1.$$

So

$$\frac{dp^*}{d\alpha} = -\frac{F_\alpha}{F_p} = -\frac{-1}{5/w + 5} = \frac{1}{5/w + 5}.$$

At $w = 1$ this is $1/10$. Direct check: from $F = 0$,

$$p\Big(\frac{5}{w} + 5\Big) = 100 + \alpha \implies p^* = \frac{100 + \alpha}{5/w + 5},$$

and the derivative with respect to α is $1/(5/w + 5)$. Same answer.

(c) $F_w = \dfrac{\partial}{\partial w}\Big(\dfrac{5p}{w}\Big) = -\dfrac{5p}{w^2}$. So

$$\frac{dp^*}{dw} = -\frac{F_w}{F_p} = \frac{5p/w^2}{5/w + 5}.$$

At $w = 1$, $p^* = 10$: $\dfrac{dp^*}{dw} = \dfrac{50}{10} = 5 > 0$. A higher wage raises marginal cost, supply shifts left, and the price rises. In the formula, the numerator $\partial Y^S/\partial w < 0$, and the sign of dp^*/dw is minus that.

(d) At $w = 1$, $\alpha = 0$, $p = 10$, $Y = X = 50$:

$$\varepsilon_{S,w} = \frac{\partial Y^S}{\partial w}\frac{w}{Y} = -\frac{5p}{w^2} \cdot \frac{w}{Y} = -\frac{50}{1} \cdot \frac{1}{50} = -1, \qquad \varepsilon_{D,w} = 0,$$

$$\varepsilon_{S,p} = \frac{5}{w} \cdot \frac{p}{Y} = 5 \cdot \frac{10}{50} = 1, \qquad \varepsilon_{D,p} = -5 \cdot \frac{p}{X} = -5 \cdot \frac{10}{50} = -1.$$

Then

$$\varepsilon_{p,w} = -\frac{\varepsilon_{S,w} - \varepsilon_{D,w}}{\varepsilon_{S,p} - \varepsilon_{D,p}} = -\frac{-1-0}{1-(-1)} = \frac{1}{2}.$$

Check with (c): $\varepsilon_{p,w} = \dfrac{dp^*}{dw} \cdot \dfrac{w}{p} = 5 \cdot \dfrac{1}{10} = \dfrac{1}{2}$. A 1% wage increase raises the price by 0.5%.

Exercise 4 (elasticities)

(a) Let $x = k\,p^{\eta}$ with $k > 0$, $\eta < 0$. Find $\varepsilon_{x,p}$ in two ways: with the definition, and by taking logs.

(b) Let $x = a - bp$ with $a, b > 0$. Find $\varepsilon_{x,p}$. At which price is it -1? Evaluate it for $X^D = 100 - 5p$ at $p = 4$, 10 and 16.

(c) With Cobb-Douglas demand $x = \alpha M/p_x$, find the own-price, cross-price and income elasticities.

(d) Empirical work says supply elasticity is 0.5 and demand elasticity is -1. A fad raises demand by 6% at every price. By how much do price and quantity change?

Solution.

(a) Definition: $\partial x/\partial p = k\eta p^{\eta-1}$, so

$$\varepsilon_{x,p} = k\eta p^{\eta-1} \cdot \frac{p}{kp^{\eta}} = \eta\,\frac{p^{\eta}}{p^{\eta}} = \eta.$$

Logs: $\ln x = \ln k + \eta \ln p$, so $\partial \ln x/\partial \ln p = \eta$. The elasticity is the same at every price. That is why this is called constant-elasticity demand.

(b) $\partial x/\partial p = -b$, so

$$\varepsilon_{x,p} = -b \cdot \frac{p}{a - bp}.$$

Set it to -1: $bp = a - bp$, so $p = a/(2b)$. At higher prices demand is elastic ($|\varepsilon| > 1$), at lower prices inelastic. The slope is the same everywhere, but the elasticity is not.

For $X^D = 100 - 5p$ ($a = 100$, $b = 5$), unit elasticity is at $p = 10$:

$$p = 4: \; -\frac{5 \cdot 4}{100 - 20} = -\frac{20}{80} = -0.25, \qquad p = 10: \; -\frac{50}{50} = -1, \qquad p = 16: \; -\frac{80}{20} = -4.$$

(c) Own price: $\dfrac{\partial x}{\partial p_x}\dfrac{p_x}{x} = -\dfrac{\alpha M}{p_x^2} \cdot \dfrac{p_x}{\alpha M/p_x} = -1$. Cross price: x does not depend on p_y, so 0. Income: $\dfrac{\alpha}{p_x} \cdot \dfrac{M}{\alpha M/p_x} = 1$. Logs give all three at once: $\ln x = \ln \alpha + \ln M - \ln p_x$.

(d) Let α be the fad, scaled so that $\varepsilon_{D,\alpha} = 1$ and $\varepsilon_{S,\alpha} = 0$, and let α rise by 6%. Then

$$\varepsilon_{p,\alpha} = -\frac{0-1}{0.5-(-1)} = \frac{1}{1.5} = \frac{2}{3}, \qquad \%\Delta p = \frac{2}{3}\cdot 6\% = 4\%.$$

Quantity from the supply side: $\%\Delta Q = \varepsilon_{S,p}\cdot\%\Delta p = 0.5\cdot 4\% = 2\%$. Check from the demand side: demand shifts out 6%, and moving along it the 4% higher price cuts it by $1\cdot 4\%$, so $6\% - 4\% = 2\%$. Both sides agree.

Exercise 5 (who pays the tax?) Use the market of Exercise 1: $Y^S = 5p$ for $p \geq 4$, and $X^D = 100 - 5p$. The government levies a tax of t per unit, collected from sellers.

(a) Find the price buyers pay, the price sellers keep, and the quantity, as functions of t.

(b) Let $t = 4$. Compute all three, the output of each firm, and tax revenue.

(c) Check $\partial p^*/\partial t$ against the elasticity formula at $t = 0$.

(d) Now the tax is collected from buyers instead. Redo (a). What changes?

(e) General linear case: $Y^S = c + dp$, $X^D = a - bp$ with $b, d > 0$. Find $\partial p^*/\partial t$. What happens when $d \to 0$, when $d \to \infty$, and when $b \to 0$?

(f) The payroll tax is split by law: half paid by the employer, half by the worker. Studies find labor supply is quite inelastic and labor demand more elastic. Who really pays?

Solution.

(a) Sellers keep $p - t$, so they supply $5(p - t)$. Equilibrium:

$$5(p-t) = 100 - 5p \implies 5p - 5t = 100 - 5p \implies 10p = 100 + 5t \implies p^* = 10 + \frac{t}{2}.$$

Sellers keep $p^* - t = 10 - \dfrac{t}{2}$. Quantity:

$$Q^* = 100 - 5\Big(10 + \frac{t}{2}\Big) = 100 - 50 - \frac{5t}{2} = 50 - \frac{5t}{2}.$$

This is valid while sellers' price is at least 4: $10 - t/2 \geq 4$, that is, $t \leq 12$.

(b) With $t = 4$: buyers pay 12, sellers keep 8, $Q^* = 50 - 10 = 40$. Each firm produces $y = 8/2 = 4$ (it reacts to the price it keeps). Check: $10\cdot 4 = 40$. Revenue $= t\,Q^* = 4\cdot 40 = 160$. Buyers pay 2 more and sellers get 2 less: they split the tax evenly.

(c) At $t = 0$: $p = 10$, $Q = 50$, so $\varepsilon_{S,p} = 5\cdot\frac{10}{50} = 1$ and $\varepsilon_{D,p} = -5\cdot\frac{10}{50} = -1$. Then

$$\frac{\partial p^*}{\partial t} = \frac{\varepsilon_{S,p}}{\varepsilon_{S,p} - \varepsilon_{D,p}} = \frac{1}{1-(-1)} = \frac{1}{2}, \qquad \frac{\partial (p^* - t)}{\partial t} = \frac{\varepsilon_{D,p}}{\varepsilon_{S,p} - \varepsilon_{D,p}} = \frac{-1}{2} = -\frac{1}{2}.$$

Same as (a): $\partial p^*/\partial t = 1/2$.

(d) Now let p be the price sellers get. Buyers pay $p + t$ and demand $100 - 5(p + t)$:

$$5p = 100 - 5(p + t) \implies 10p = 100 - 5t \implies p^* = 10 - \frac{t}{2}.$$

Buyers pay $p^* + t = 10 + t/2$, and $Q^* = 5(10 - t/2) = 50 - 5t/2$. Exactly as in (a). Only the label on the price changes. The legal side of the tax is irrelevant.

(e) Equilibrium with the tax:

$$c + d(p - t) = a - bp \implies (b + d)\, p = a - c + dt \implies p^* = \frac{a - c + dt}{b + d}, \qquad \frac{\partial p^*}{\partial t} = \frac{d}{b + d}.$$

- $d \to 0$ (vertical supply, e.g. land): $\partial p^*/\partial t \to 0$. Buyers pay nothing. Sellers pay the whole tax.
- $d \to \infty$ (flat supply, constant returns): $\partial p^*/\partial t \to 1$. Buyers pay everything.
- $b \to 0$ (vertical demand): $\partial p^*/\partial t \to 1$. Buyers pay everything.

The less elastic side always pays more.

(f) By (d), the legal split does not matter. Workers are the sellers of labor. If their supply is less elastic than firms' demand, the wage workers take home falls by most of the tax. Workers bear most of the payroll tax, including the employer's "half".